

Homoglyph Detection Tool

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Proof of Concept (PoC) Report

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Project Summary:

This tool is a web-based Homoglyph Detection system that identifies visually similar Unicode characters (homoglyphs) in user input. It is designed to detect phishing-style attacks in domain names, email addresses, or any text where malicious characters may impersonate legitimate ones.

Technologies Used:

- JavaScript
 - HTML5
 - Unicode (Homoglyph Mapping)
 - JSON for extensibility
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Problem Statement:

Attackers often use Unicode characters that resemble Latin letters to trick users into clicking fake URLs or reading malicious text. These characters are called homoglyphs. This tool aims to detect and flag such characters in real-time from text input.

Objective:

To implement a tool that:

- Allows live input of domain, email, or text

- Detects visually deceptive characters (homoglyphs)
 - Highlights the suspicious characters
 - Displays explanations and suggested base letters
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Tool Features:

- Homoglyph mapping using Unicode
 - Real-time input detection
 - Custom homoglyph list from JSON
 - Type-specific detection (email, domain, text)
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PoC Proofs:

Users can choose Domain, Email, or Text input types for Homoglyph Detection.

Input Handling:

```
javascript

let inputField = document.getElementById("inputText");
inputField.addEventListener("input", checkHomoglyphs);
```

This selects the input field and listens for real-time input, calling `checkHomoglyphs()` when the user types.

Homoglyph Detection:

```
javascript

let input = inputField.value;
for (let i = 0; i < input.length; i++) {
  let char = input[i];
  for (let base in homoglyphs) {
    if (homoglyphs[base].includes(char)) {
      found.push(`${char}` at position ${i + 1} (looks like `${base}`));
    }
  }
}
```

This code checks each character for homoglyphs and adds suspicious ones to the `found` array.

Displaying the Result:

```
javascript

result.innerHTML = "message here";
```

The page dynamically shows whether homoglyphs were detected using `innerHTML`.

Tool Execution Flow:

- User types input → captured via `document.getElementById("inputText").value`
- Input triggers `checkHomoglyphs()` using `addEventListener("input")`
- Input scanned for suspicious homoglyphs → flagged characters added to `found[ ]`
- Output shown using `innerHTML`

- Optional: highlight suspicious characters with `<span style="color:red">...</span>`

Conclusion:

- The tool detects visually deceptive Unicode characters (homoglyphs) in real-time.
- Built using JavaScript, it runs directly in the browser with no installation.
- Capable of scanning domain names, emails, or any text input for phishing attempts.
- Uses a customizable `homoglyphs.json` file for easy extension.
- Enhances security awareness and supports anti-phishing efforts.
- Can be extended into browser extensions or integrated into email filters.
- Lightweight and easy to use — no external libraries required.